

Environmental Chemical Burden in Differentiated Thyroid Cancer

Human Thyroid Tumor Exposomics

- Recent increases in differentiated thyroid cancer (DTC) incidence.



- Potential contribution of environmental chemicals in epidemiologic trends.

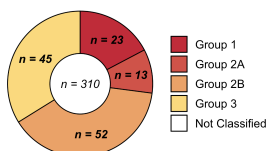


- GC-MS environmental surveillance study of 710 xenobiotic compounds n=60 DTC samples and n=8 cadaver thyroids.

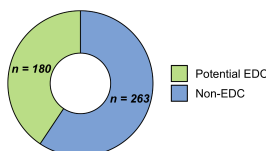


Environmental Chemical Surveillance

IARC Carcinogen Class

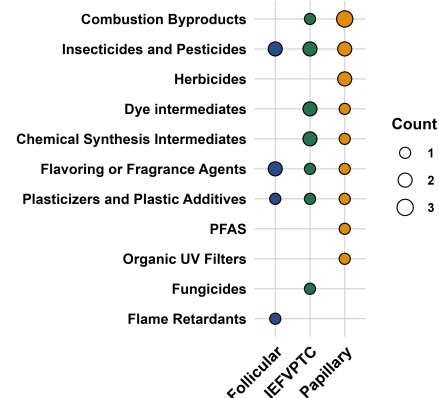


Endocrine Disrupting Chemicals



- 443 chemicals annotated in tumors, spanning insecticides, herbicides, industrial chemicals, plasticizers, flame retardants, flavoring & fragrance agents.
- Substantial portion of chemicals were EDCs or IARC1 carcinogens.

Variant-Specific Chemical Burden



- 30 identified chemicals differed between variants; papillary thyroid carcinoma appeared most burdened, with higher levels of multiple combustion byproducts, insecticides/pesticides, and herbicides.

Conclusion: This study provides evidence of a heavy environmental chemical burden in DTC; specific organic pollutants could potentially influence type-specific carcinogenesis.

Reference: Preston JD, Liang Y, Yamashita TS, et al. Environmental Chemical Burden in Differentiated Thyroid Cancer. *Environment International*. 2026.

Figure 1

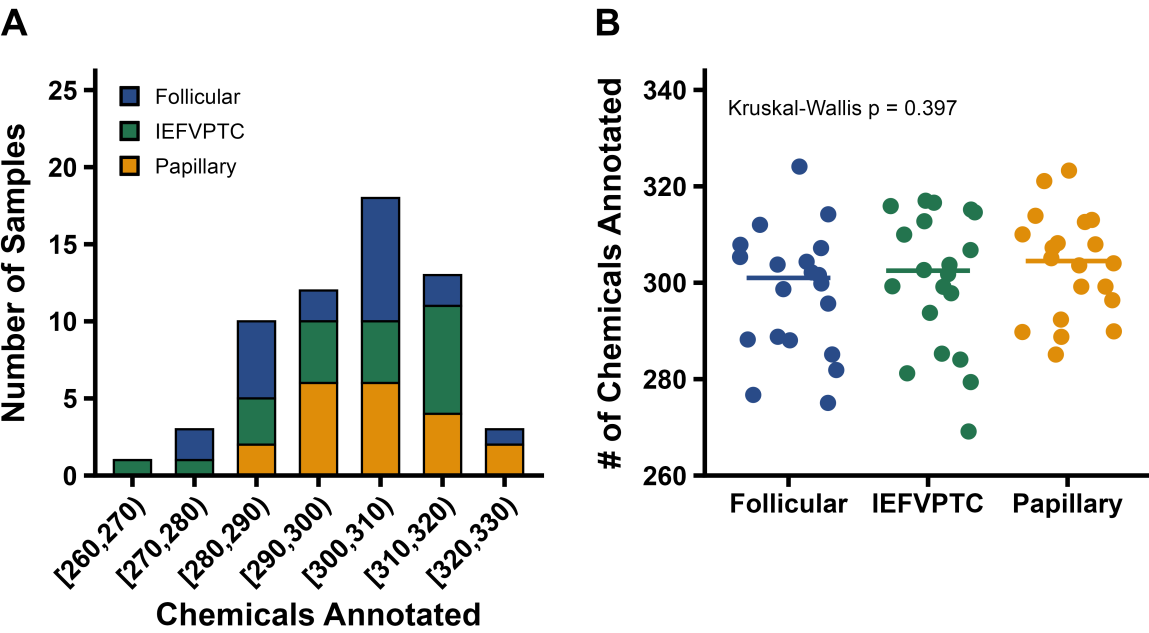
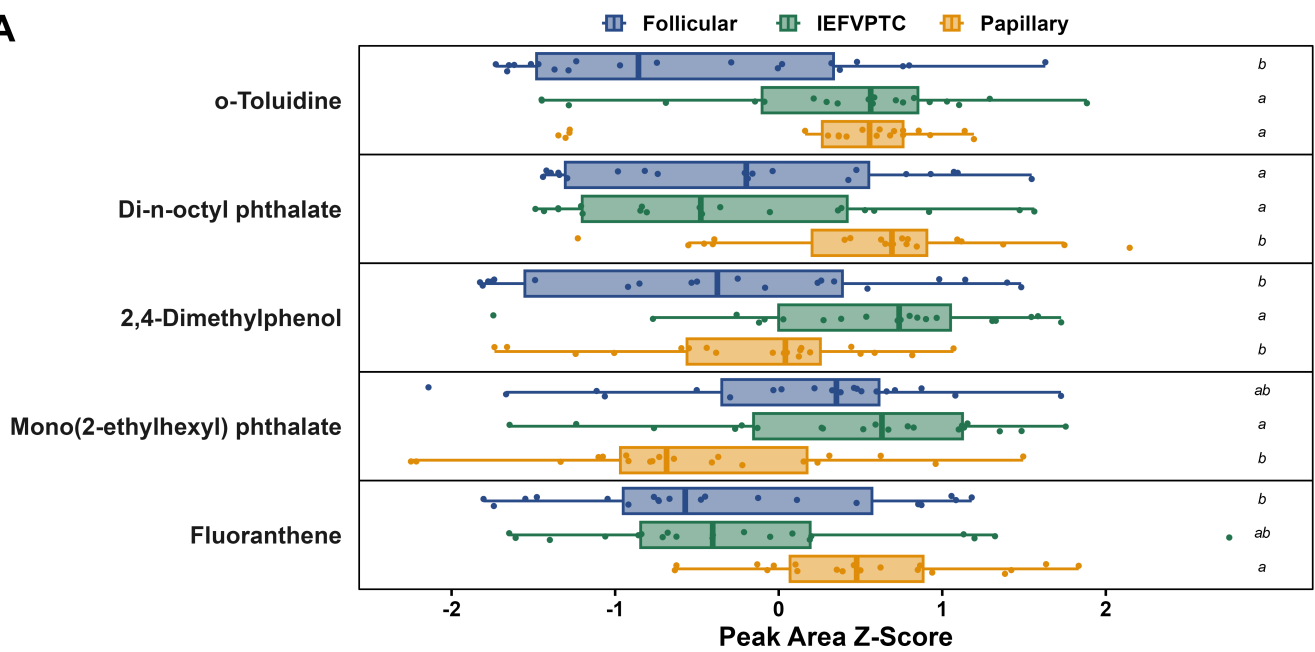


Figure 2

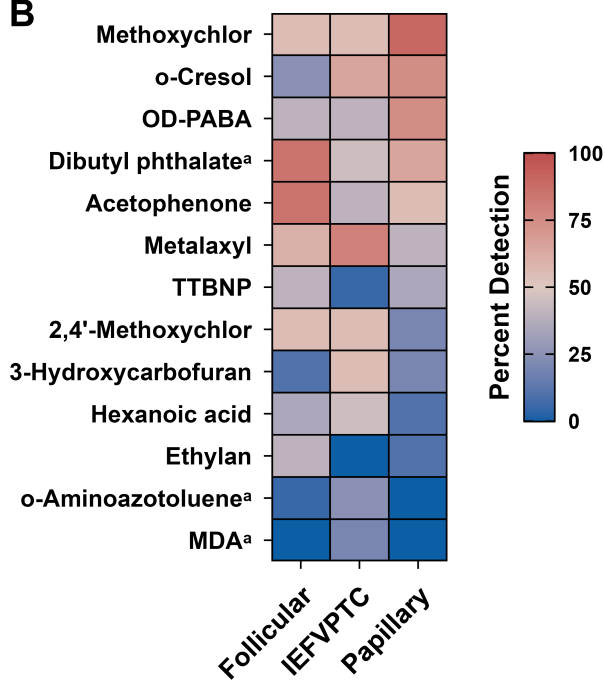


Figure 3

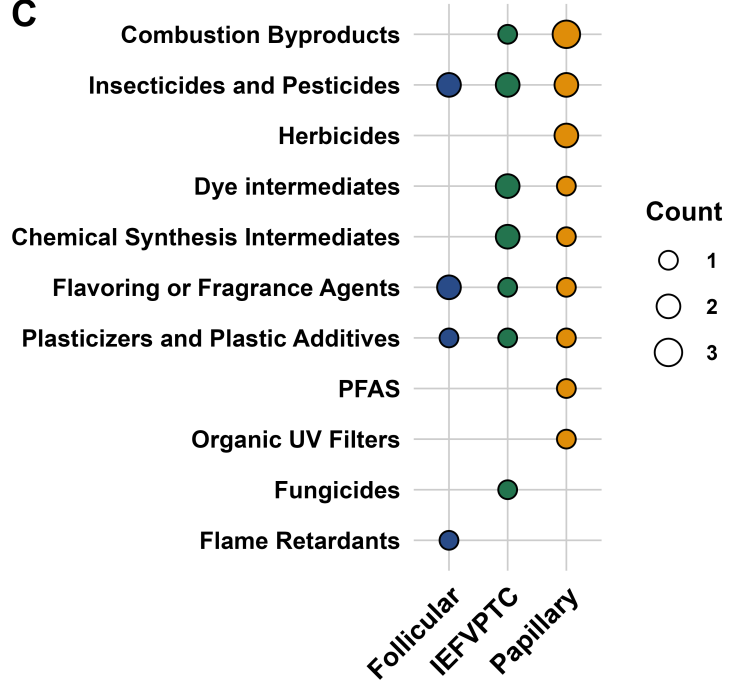
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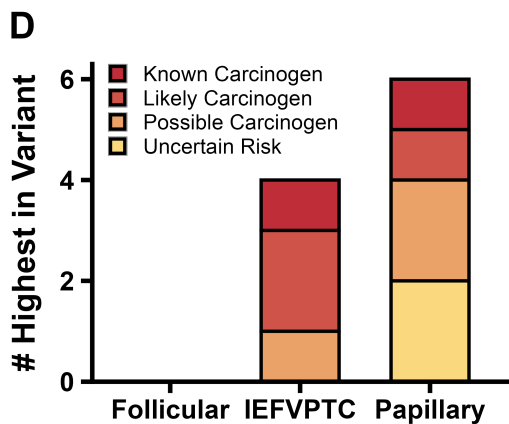
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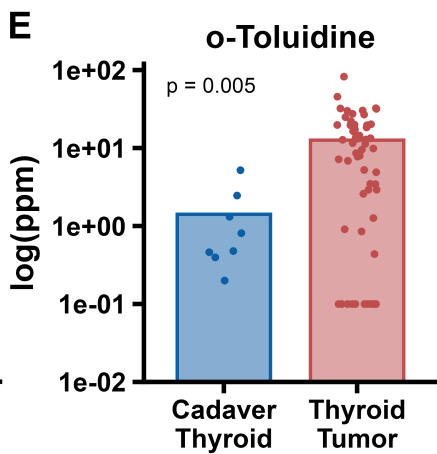
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